

ECO 344 - Applied Econometrics I

Chapter 1

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- 1 What is Econometrics?
- 2 Data & its Types
- 3 Empirical Analysis
- 4 Causal Inference
- 5 Summary & Discussion

What is Econometrics?

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Data & its Types

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 - Non-Experimental Data - Observational Data like Census Data etc.

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- Difference between Pooled Cross Sectional & Panel Data?

Empirical Analysis

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- Finally the last step is to report your findings in an organized format.

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- Random Data - The data that is collected from a sample of random subjects.

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 - The effect of COVID on your school learning?

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- Random Vs Non-Random Vs Quasi-Random Data
- Causal Inference (Ceteris Paribus, Potential Outcomes, Counterfactual)

Economic Model of Crime

$$y = f(x_1, x_2, x_3, x_4, x_5, x_6, x_7)$$

where;

$y = \text{HoursSpentinCriminalActivity}$

$x_1 = \text{Wageforanhourspentincriminalactivity}$

$x_2 = \text{Hourlywageinlegalemployment}$

$x_3 = \text{Incomeotherthanfromcrime/employment}$

$x_4 = \text{Probabilityofgettingcaught}$

$x_5 = \text{Probabilityofbeingconvictedifcaught}$

$x_6 = \text{ExpectedSentenceifconvicted}$

$x_7 = \text{Age}$

Job Training & Worker Productivity

$$wage = f(education, experience, training)$$

$$wage = \beta_0 + \beta_1 education + \beta_2 experience + \beta_3 training + \mu$$

Data Types

obsno	wage	educ	exper	female	married
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
525	11.56	16	5	0	1
526	3.50	14	5	1	0

Data Types

obsno	country	gpcrgdp	govcons60	second60
1	Argentina	0.89	9	32
2	Austria	3.32	16	50
3	Belgium	2.56	13	69
4	Bolivia	1.24	18	12
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
61	Zimbabwe	2.30	17	6

Data Types

obsno	year	avgmin	avgcov	prunemp	prgnp
1	1950	0.20	20.1	15.4	878.7
2	1951	0.21	20.7	16.0	925.0
3	1952	0.23	22.6	14.8	1015.9
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
37	1986	3.35	58.1	18.9	4281.6
38	1987	3.35	58.2	16.8	4496.7

Data Types

obsno	year	hprice	proptax	sqrft	bdrms	bthrms
1	1993	85,500	42	1600	3	2.0
2	1993	67,300	36	1440	3	2.5
3	1993	134,000	38	2000	4	2.5
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
250	1993	243,600	41	2600	4	3.0
251	1995	65,000	16	1250	2	1.0
252	1995	182,400	20	2200	4	2.0
253	1995	97,500	15	1540	3	2.0
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
520	1995	57,200	16	1100	2	1.5

Data Types

obsno	city	year	murders	population	unem	police
1	1	1986	5	350,000	8.7	440
2	1	1990	8	359,200	7.2	471
3	2	1986	2	64,300	5.4	75
4	2	1990	1	65,100	5.5	75
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
297	149	1986	10	260,700	9.6	286
298	149	1990	6	245,000	9.8	334
299	150	1986	25	543,000	4.3	520
300	150	1990	32	546,200	5.2	493

Question

- 1 Suppose that you are asked to conduct a study to determine whether smaller class sizes lead to improved student performance of fourth graders.
 - (i) If you could conduct any experiment you want, what would you do? Be specific.
 - (ii) More realistically, suppose you can collect observational data on several thousand fourth graders in a given state. You can obtain the size of their fourth-grade class and a standardized test score taken at the end of fourth grade. Why might you expect a negative correlation between class size and test score?
 - (iii) Would a negative correlation necessarily show that smaller class sizes cause better performance? Explain.